

Gate 0 Technical Report: Auditing the Observer-Entropy to Horizon-Entropy Bridge in Physics v1.0

Technical Report v1.0 – Frozen

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Audit target. *An Information-Theoretic Framework for Emergent Spacetime Dynamics, Horizon Thermodynamics, and Testable Cosmological Signatures* (Physics v1.0).

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Purpose statement. This report audits an open problem Physics v1.0 explicitly identified. It is not a critique of v1.0; it is a systematic investigation of an uncertainty v1.0 left open.

Abstract

Physics v1.0 defines an observer reduced state by an assumed observer–environment Hilbert-space factorization and uses its von Neumann entropy in bits, H_{obs} , as an information-theoretic state variable. The same paper explicitly states that applying a causal-horizon area–entropy scale to this subsystem is a structural domain extension rather than a derivation, and that the microphysical observer–horizon mapping remains open. Gate 0 asks whether established entanglement and semiclassical-gravity results close that bridge without adding assumptions absent from v1.0. We reconstruct two area-law calculations (Bombelli–Koul–Lee–Sorkin and Srednicki), audit the Susskind–Uglum renormalization connection and its higher-curvature extensions, distinguish matter entanglement entropy, gravitational entropy, and generalized entropy, and finally test the original H_{obs} definition against the structural requirements extracted from those results. The audit finds that the known bridges rely on a geometrically specified bipartition, regulated local quantum-field structure, and coupling to a gravitational effective action. None is supplied by the original definition of H_{obs} . Gate 0 therefore receives the frozen verdict **UNRESOLVED IN GENERALITY**. This is not a proof of impossibility. It is a stop condition: Gate 1 is not entered from the current assumptions.

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1 Scope, logic, and verdict vocabulary

The report is divided into two logically separated tracks. **Gate 0A** examines geometric entanglement and semiclassical black-hole constructions. **Gate 0B** asks whether those results can be transferred to the specific observer entropy defined in Physics v1.0. The following firewall is binding throughout:

A result established in Gate 0A may not be used in Gate 0B unless the connecting proposition is independently

(1)

Verdicts are used as follows. **PASS** means that the stated result follows within the explicitly delimited domain. **CONDITIONAL** means that the bridge works only after substantive additional assumptions or within restricted model/renormalization conditions. **FAIL** is reserved for an actual incompatibility or contradiction. **UNRESOLVED IN GENERALITY** means that neither the literature reviewed nor the assumptions of Physics v1.0 establish the general bridge; it is not equivalent to FAIL.

The permanent entropy notation is

$$S_{\text{ent}}^{\text{matter}} \neq S_{\text{grav}} \neq S_{\text{gen}} \quad (2)$$

unless a particular framework supplies and justifies a relation among them. Equality is never inferred from dimensional similarity or from the fact that each quantity is called "entropy."

2 Physics v1.0 definitions under audit

This section quotes the frozen source passages verbatim. These statements define the object whose possible gravitational interpretation is being audited.

2.1 Observer–environment factorization (source lines 50–54)

"In the present framework, we additionally **assume** that the relevant Hilbert space admits an observer–environment tensor decomposition,

$$\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{obs}} \otimes \mathcal{H}_{\text{env}}.$$

This factorization is a structural modeling assumption; it is not derived from the Wheeler–DeWitt constraint."

[ESTABLISHED FROM v1.0 DEFINITION] The tensor factorization is therefore an explicit model assumption. No spatial, causal, or horizon localization is part of the definition itself.

2.2 Reduced density operator (source lines 60–65)

"Standard quantum mechanics then defines the observer reduced density operator by the partial trace,

$$\rho_{\text{obs}} = \text{Tr}_{\text{env}}(|\Psi\rangle\langle\Psi|) = \sum_j \langle\chi_j|\Psi\rangle\langle\Psi|\chi_j\rangle.$$

"

[ESTABLISHED] Once a tensor factorization and global state are given, the partial trace is standard quantum mechanics. No gravitational interpretation follows from the partial trace alone.

2.3 H_{obs} definition (source lines 75–81)

“For the dynamical model below, we define the same von Neumann entropy expressed in bits,

$$H_{\text{obs}} \equiv \frac{S_{\text{vN}}}{k_B \ln 2} = -\text{Tr}(\rho_{\text{obs}} \log_2 \rho_{\text{obs}}).$$

Here H_{obs} is dimensionless in physical dimensions, with its numerical value expressed in bits. Within the proposed framework it is used as an information-theoretic state variable associated with observer–environment correlations. Identifying this variable with macroscopic organizational or self-modeling capacity is an explicit modeling hypothesis rather than a consequence of the Wheeler–DeWitt constraint or the partial-trace construction.”

[ESTABLISHED FROM v1.0 DEFINITION] Thus $k_B \ln 2 H_{\text{obs}}$ is the von Neumann entropy of the chosen observer factor, not by definition a horizon entropy.

2.4 Horizon boundary condition and explicit caveat (source lines 143–149)

“For the present phenomenological model, we introduce the boundary condition

$$H_{\text{obs}} \leq H_{\text{max}}, \quad H_{\text{max}} \equiv \frac{A}{4\ell_P^2 \ln 2},$$

for a specified causal-horizon area A . **Applying the horizon area–entropy scale to this observer subsystem is a structural domain extension of the model, not a derivation that identifies H_{obs} with microscopic horizon entropy. The microphysical mapping between observer degrees of freedom and gravitational horizon degrees of freedom remains open.**”

[ESTABLISHED FROM v1.0 DEFINITION] The inequality is a phenomenological boundary condition. It cannot be promoted to a microscopic identity without an additional bridge.

2.5 Conclusion statement (source lines 201–202)

“Standard quantum reduction supplies ρ_{obs} once an observer–environment factorization is assumed; von Neumann entropy is kept distinct from its bit-normalized representation; Landauer’s principle is restricted to logically irreversible erasure; and the proposed dynamical equations are dimensionally closed with explicit coefficient units. Horizon thermodynamics and Jacobson’s recovery of the Einstein equation provide established gravitational context, **while the identification of observer information with horizon degrees of freedom remains open.**”

The Gate 0 question is consequently narrow: *do established geometric-entanglement and semiclassical-gravity constructions close this explicitly open mapping under the original v1.0 assumptions?*

3 Arrow 1A: Srednicki (1993) – geometric vacuum entanglement

3.1 Model and reduction

[ESTABLISHED FROM LITERATURE] Srednicki studies the ground state of a free massless scalar field in 3 + 1 dimensional flat spacetime, traces the degrees of freedom inside an imaginary sphere, and finds an entropy proportional to the sphere’s area rather than its volume [1].



After discretizing the radial direction with short-distance spacing a , a spherical-harmonic decomposition reduces the free field to sectors of coupled harmonic oscillators. A generic sector has quadratic form

$$H = \frac{1}{2} p^T p + \frac{1}{2} x^T K x, \quad (3)$$

with positive coupling matrix K . Its ground state is Gaussian,

$$\Psi_0(x) = \mathcal{N} \exp \left[-\frac{1}{2} x^T \Omega x \right], \quad \Omega = K^{1/2}. \quad (4)$$

Splitting the lattice coordinates into traced and retained variables, $x = (x_{\text{in}}, x_{\text{out}})$, gives

$$\rho_{\text{out}}(x, x') = \int dx_{\text{in}} \Psi_0(x_{\text{in}}, x) \Psi_0^*(x_{\text{in}}, x'). \quad (5)$$

[SHARED TRACTABILITY ASSUMPTION] The global state is Gaussian, the bipartition acts linearly on oscillator variables, and integration over a subset of Gaussian variables preserves Gaussian form. This makes the reduced-state entropy computable from Gaussian spectral/covariance data. It is not a universal property of interacting or non-Gaussian theories.

3.2 Leading area structure

[DERIVED] Locality and short-distance correlations make the leading contribution sensitive to modes near the entangling surface. In $3 + 1$ dimensions the number of UV cells intersecting a surface of area A scales as

$$N_{\partial} \sim \frac{A}{a^2}. \quad (6)$$

The regulated leading entropy therefore has the schematic structure

$$S_{\text{ent}}^{\text{matter}} \sim k_B C_S \frac{A}{a^2} = k_B C_S M^2 A, \quad M \sim a^{-1}. \quad (7)$$

[REGULATOR DEPENDENT] The leading coefficient C_S depends on the UV prescription; the continuum limit of the bare leading term diverges as $a \rightarrow 0$. The area scaling is the robust structural result of this controlled calculation, not a derivation of the Bekenstein–Hawking coefficient.

$$a \sim \ell_P \Rightarrow S = \mathcal{O}(k_B A / \ell_P^2) \text{ is an order-of-magnitude estimate, not a derivation of } 1/4. \quad (8)$$

3.3 Arrow 1A verdict

$$\boxed{\text{PASS} - \text{restricted domain}} \quad (9)$$

For a regulated free massless scalar vacuum in $3 + 1$ flat spacetime with a geometrically defined spherical bipartition, the leading entanglement entropy scales with the separating area. This does not establish $S_{\text{ent}}^{\text{matter}} = S_{\text{BH}}$ and says nothing by itself about the v1.0 observer/environment split.

4 Arrow 1B: Bombelli–Koul–Lee–Sorkin (1986)

4.1 Black-hole-motivated reduced state

[ESTABLISHED FROM LITERATURE] BKLS associate an effective density matrix with quantum-field degrees of freedom in a black-hole background and interpret its statistical entropy

as a possible contribution to total black-hole entropy. Their explicit evaluation uses a simplified coupled-oscillator model and yields an area-proportional contribution [2].

The computational core again has a quadratic Hamiltonian. To avoid confusing a kinetic matrix with Newton's constant, write

$$H = \frac{1}{2}p^T K_p^{-1} p + \frac{1}{2}q^T V q, \tag{10}$$

with Gaussian ground state

$$\Psi_0(q) \propto \exp\left(-\frac{1}{2}q^T \Omega q\right). \tag{11}$$

A split $q = (q_{\text{in}}, q_{\text{out}})$ and partial trace over inaccessible variables gives a Gaussian reduced state. The same Gaussian tractability condition noted in Arrow 1A therefore applies.

4.2 Area scaling and cutoff interpretation

[DERIVED] For a local lattice with spacing a , short-range correlations straddling the interface give schematically

$$N_\partial \sim \frac{A_H}{a^2}, \quad S_{\text{ent}}^{\text{matter}} \sim k_B C_B \frac{A_H}{a^2}. \tag{12}$$

Here A_H is a black-hole horizon area rather than an artificial sphere in Minkowski space. The shared A/a^2 structure does not make the two physical partitions identical.

[MODEL DEPENDENT] BKLS's abstract describes the simplified contribution as expected to be finite and area-proportional. Later continuum and one-loop analyses make the UV-divergence/renormalization structure explicit. This report does not retroactively attribute those later renormalization results to BKLS.

4.3 Srednicki–BKLS comparison

Property	Srednicki 1993	BKLS 1986
System	Free massless scalar vacuum	Scalar-field-inspired coupled-oscillator model in black-hole context
Background	Minkowski spacetime	Black-hole motivated background/partition
Partition	Inside/outside an imaginary sphere	Inaccessible/accessible degrees associated with horizon
Boundary meaning	Artificial spatial entangling surface	Physical causal-horizon motivation
Reduction	Partial trace over one spatial region	Partial trace over inaccessible variables
UV implementation	Radial lattice	Spatial oscillator lattice/cutoff in simplified model
Leading structure	$S \sim k_B C_S A/a^2$	$S \sim k_B C_B A_H/a^2$ schematically
Coefficient	Regulator dependent	Cutoff/model dependent
Newton's G derived?	No	No
$S = S_{\text{BH}}$ established?	No	No
Relation to $H_{\text{obs}}^{v1.0}$	None established	None established

The common structural result is therefore

local short-distance correlations + geometric division + partial trace + UV regulation $\Rightarrow$ leading boundary-area

(13)

in these controlled free/Gaussian constructions. Universality of the coefficient is not established.

4.4 Arrow 1B verdict

PASS – model/domain conditional

 (14)

The horizon-motivated reduced-state construction supports an area-proportional entropy contribution in its simplified setting. It does not derive the Bekenstein–Hawking coefficient and does not connect the v1.0 observer factor to the horizon factor.

5 Arrow 2: Susskind–Uglum renormalization and its limits

5.1 Matter entropy divergence

[ESTABLISHED FROM LITERATURE] Susskind and Uglum study an eternal Schwarzschild black hole in the infinite-mass limit. For a free scalar field on a fixed black-hole background they find the entropy per unit area to be quadratically divergent near the horizon, and relate the quantum correction to entropy per area to the quantum correction to the gravitational coupling [3].

In four dimensions the leading matter contribution may be written schematically as

$$S_{\text{matter}}^{\text{div}} = k_B C_{\text{field,reg}} \frac{A_H}{\epsilon^2} + \cdots, \quad (15)$$

where ϵ is a short-distance regulator. The coefficient depends on field content and regularization details. The bare matter contribution therefore does not furnish a finite universal S_{BH} .

5.2 Effective gravitational action

Matter loops also renormalize the gravitational effective action. Schematically,

$$I_{\text{eff}} = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G_{\text{bare}}} + \delta \mathcal{L}_{\text{matter}} + c_{1,\text{bare}} R^2 + c_{2,\text{bare}} R_{\mu\nu} R^{\mu\nu} + \cdots \right]. \quad (16)$$

Renormalization reorganizes the divergent parameters, for example

$$\frac{1}{G_{\text{ren}}} = \frac{1}{G_{\text{bare}}} + \delta \left(\frac{1}{G} \right), \quad (17)$$

with analogous relations for higher-curvature couplings.

[ESTABLISHED FROM LITERATURE] Fursaev and Solodukhin show at one loop that black-hole entropy divergences are eliminated by the standard renormalization of the gravitational constant together with coefficients of R^2 -type terms in the effective action; those higher-curvature terms generate entropy contributions depending on horizon geometry [4].

The structurally earned statement is therefore

$S_{\text{matter}}^{\text{div}} \longleftrightarrow$ UV renormalization of gravitational effective-action couplings

 (18)

within the controlled semiclassical setting. This is stronger than setting $\epsilon = \ell_P$ by hand, but weaker than a universal identity $S_{\text{ent}}^{\text{matter}} = A/(4G)$.

5.3 Field content and nonminimal coupling

[ESTABLISHED FROM LITERATURE] Solodukhin extends the renormalization statement to nonminimally coupled scalar matter and explicitly discusses the relation between entanglement



and thermodynamic entropies [5]. Thus field type and couplings are not dispensable bookkeeping: they affect the matter divergence and the renormalization problem.

For a theory containing several species one may write schematically

$$S_{\text{matter}}^{\text{div}} = S_{\text{div}}(N_s, N_f, N_v, m_i, \xi_i, \dots; \epsilon). \quad (19)$$

Bare matter entropy is field-content dependent. The corresponding loops also renormalize gravitational couplings, so field dependence of the bare entropy alone does not prove that the final renormalized gravitational entropy retains arbitrary species dependence. What it does prove is that no field-independent identification can be asserted before the renormalization structure and theory are specified.

5.4 Arrow 2 verdict

CONDITIONAL (20)

In controlled black-hole/semi-classical settings, UV-divergent matter contributions are tied to renormalization of gravitational effective-action couplings. The result is conditional on the field theory, regulator/renormalization framework, background, and gravitational effective action. It does not establish a regulator-independent universal finite remainder equal to the von Neumann entropy of an arbitrary subsystem.

6 Arrow 3: Entropy ontology after renormalization

Arrow 3 asks what quantity is being calculated after the renormalization machinery is introduced. Three objects must remain distinct.

6.1 Matter entanglement entropy

For a geometrically specified field-theory partition,

$$S_{\text{ent}}^{\text{matter}} = -k_B \text{Tr}(\rho_A \ln \rho_A). \quad (21)$$

It is a reduced-state entropy of quantum matter degrees of freedom. In continuum QFT its bare value generally has UV divergences associated with the entangling surface.

6.2 Gravitational entropy

S_{grav} denotes entropy obtained from the gravitational action/effective action. In pure Einstein gravity its stationary black-hole contribution is the Bekenstein–Hawking area term,

$$S_{\text{BH}} = \frac{k_B A_H}{4\ell_P^2} = \frac{k_B c^3 A_H}{4\hbar G}. \quad (22)$$

For higher-curvature effective actions, the gravitational entropy generally contains additional geometric terms; Fursaev–Solodukhin’s one-loop analysis makes this necessary for the renormalization problem [4].

6.3 Generalized entropy

A semiclassical generalized-entropy construction has the schematic form

$$S_{\text{gen}} = S_{\text{grav}} + S_{\text{out}}, \quad (23)$$

where S_{out} is the appropriate quantum-field entropy outside (or on a specified side of) the surface, subject to a consistent renormalization prescription. This is a combined quantity; it is not simply a renaming of either term.

Consequently,

$$S_{\text{ent}}^{\text{matter}} \neq S_{\text{grav}} \neq S_{\text{gen}} \quad (24)$$

by definition and historical role unless a specific theory supplies a further equality or relation. Dimensional analysis cannot provide such an identification: all three may have entropy units while encoding different state, geometric, and renormalization data.

[OPEN] Susskind–Uglum’s field-theory renormalization relation does not, by itself, constitute a general microscopic counting theorem for the classical Bekenstein–Hawking entropy. Their paper separately discusses superstring configurations as a candidate microscopic origin for the classical contribution [3]. This separation is evidence against silently identifying the ordinary matter reduced-state entropy with the full microscopic gravitational entropy.

6.4 Arrow 3 verdict

$$\boxed{\text{CONDITIONAL} / \text{interpretation-dependent}} \quad (25)$$

The renormalized effective-action entropy is a gravitational/thermodynamic quantity in the relevant semiclassical construction, while matter entanglement is a reduced-state quantity and generalized entropy combines gravitational and exterior matter terms. Their relations are framework dependent. The reviewed results do not establish a universal ontology in which all three are the same microscopic entropy.

7 Arrow 4: Necessary-conditions audit of H_{obs}

Arrow 4 does not attempt to prove $H_{\text{obs}} = S_{\text{BH}}/(k_B \ln 2)$. It asks what structural properties the v1.0 object would need to possess for Arrows 1–3 even to apply.

7.1 Condition A: geometric localization

Arrows 1A and 1B use a geometric or horizon-motivated bipartition that isolates correlations across a codimension-two interface. To transfer that result one would need, at minimum, a proposition such as

$$\mathcal{H}_{\text{obs}} \otimes \mathcal{H}_{\text{env}} \overset{?}{\longleftrightarrow} \mathcal{H}_{\text{inside}} \otimes \mathcal{H}_{\text{outside}} \quad (26)$$

with a defined entangling/horizon surface Σ and area $A(\Sigma)$.

[NOT ESTABLISHED FOR v1.0] Physics v1.0 defines an observer/environment tensor factorization but does not identify either factor with spatial support, a causal wedge, an apparent/event horizon, or local QFT degrees of freedom. Therefore no Σ_{obs} or A_{obs} follows from the definition.

$$\boxed{\text{Condition A: NOT SATISFIED BY THE v1.0 DEFINITION ALONE}} \quad (27)$$

7.2 Condition B: regulated local-QFT/UV structure

The area-law and renormalization calculations require specified fields, dynamics, locality, a UV prescription, and a continuum/regulator limit. Merely writing

$$H_{\text{obs}} = -\text{Tr}(\rho_{\text{obs}} \log_2 \rho_{\text{obs}})$$
 (28)

does not imply

$$S \sim C \frac{A}{\epsilon^2}.$$
 (29)

[NOT ESTABLISHED FOR v1.0] The v1.0 definition does not specify the field content of $\mathcal{H}_{\text{obs}}$, a local QFT Hamiltonian, UV cutoff ϵ , renormalization scale, or regulator. Therefore the divergence structure required by Arrow 2 cannot be inferred.

Condition B: NOT SATISFIED BY THE v1.0 DEFINITION ALONE

 (30)

7.3 Condition C: gravitational effective-action coupling

Arrow 2 relates matter UV divergences to renormalization of G and higher-curvature couplings. A transferable object therefore needs a specified role in a gravitational effective action or a derived map into quantities that renormalize it.

[NOT ESTABLISHED FOR v1.0] The definition of H_{obs} contains no dependence of the form

$$H_{\text{obs}}[g_{\mu\nu}, G, c_i, \mu, \dots]$$
 (31)

and v1.0 supplies no term whose UV divergence can be matched to $\delta(1/G)$ or to higher-curvature counterterms.

Condition C: NOT SATISFIED BY THE v1.0 DEFINITION ALONE

 (32)

7.4 Three-condition diagnostic

Requirement extracted from known bridge	Required in reviewed constructions	Established for v1.0 H_{obs} ?
Geometrically defined bipartition	Yes	No
Local regulated QFT / UV structure	Yes	No
Coupling to gravitational effective-action terms	Yes	No

Here “No” means *not established by the frozen definition and assumptions*, not “mathematically impossible under every extension.” This distinction is central:

not established $\neq$ proved impossible.

 (33)



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7.5 Arrow 4 verdict

UNRESOLVED IN GENERALITY

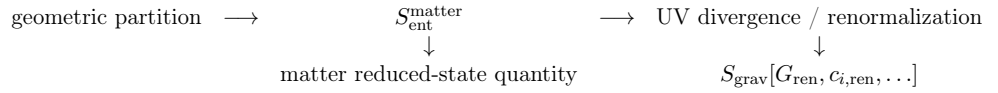
(34)

The original v1.0 definition of H_{obs} does not itself provide the geometric bipartition, regulated local-field structure, or gravitational-coupling dependence used by the known entanglement-to-gravity constructions examined in Gate 0A. Consequently those results cannot presently be transferred to H_{obs} without additional assumptions.

This result is consistent with, rather than corrective of, Physics v1.0: v1.0 explicitly states that the observer–horizon microphysical mapping remains open.

8 Dependency graph and firewall

The logical dependencies established by Gate 0 can be represented as



FIREWALL: no observer-to-geometric identification established

$$H_{\text{obs}}^{v1.0} \xrightarrow{?} S_{\text{ent}}^{\text{matter}} / (k_B \ln 2) \xrightarrow{?} S_{\text{grav}} / (k_B \ln 2) \quad (35)$$

The first row is Gate 0A and is restricted to the settings actually studied. The bottom row is Gate 0B. No result in the top row is allowed to cross the firewall merely because H_{obs} is a von Neumann entropy.

A more detailed dependency list is:

1. **[ASSUMPTION]** choose a geometric/local QFT partition and a state;
2. **[ESTABLISHED FROM LITERATURE]** in controlled free/Gaussian examples, the reduced-state entropy has leading area scaling;
3. **[REGULATOR DEPENDENT]** the leading bare term is cutoff dependent/divergent in continuum QFT;
4. **[ESTABLISHED FROM LITERATURE]** in semiclassical black-hole settings, the divergence structure is tied to renormalization of gravitational effective-action couplings;
5. **[DERIVED]** this yields a renormalized gravitational entropy construction, not a universal proof that arbitrary reduced-state entropy equals microscopic horizon entropy;
6. **[OPEN]** connecting the v1.0 observer factor to the geometric/QFT factor remains an independent proposition.

9 Frozen verdict matrix

10 Limitations and caveats

1. **This is not a no-go theorem.** Gate 0 does not prove that no future microscopic theory can relate an observer subsystem to a horizon subsystem. It shows that the reviewed bridges do not establish that relation from the assumptions currently present in Physics v1.0.
2. **The literature audit is deliberately scoped.** It reconstructs the specific historical chain needed to test the v1.0 bridge: geometric entanglement area laws, black-hole-motivated reduced states, and semiclassical renormalization. It is not a comprehensive review of holography, algebraic

Stage	Frozen verdict	Limiting statement
Arrow 1A – Srednicki	PASS – restricted domain	Regulated free massless scalar vacuum with geometric spherical partition exhibits leading area scaling; coefficient is regulator dependent.
Arrow 1B – BKLS	PASS – model/domain conditional	Horizon-motivated Gaussian reduced-state model supports area proportionality; no derivation of universal BH coefficient or v1.0 mapping.
Arrow 2 – SU / effective-action renormalization	CONDITIONAL	Matter entropy divergences are related to renormalization of gravitational couplings in controlled settings; higher-curvature and field-content details matter.
Arrow 3 – entropy ontology	CONDITIONAL / interpretation-dependent	Matter entanglement, gravitational entropy, and generalized entropy remain distinct objects unless a specified theory relates them.
Arrow 4 – H_{obs} bridge	UNRESOLVED IN GENERALITY	v1.0 does not supply the geometric, regulated-QFT, or gravitational-effective-action structures required to transfer Arrows 1–3.

- QFT, quantum extremal surfaces, induced gravity, string microstate counting, or all modern approaches to gravitational entropy.
- Gaussian examples are not universal.** The explicit Arrow 1 reconstructions rely on free/Gaussian structure for tractability. Interacting QFTs can retain area-law divergence structure but require more sophisticated methods; no such general result is imported into v1.0 without a specified theory.
 - Regulator and renormalization statements are framework sensitive.** Bare entanglement entropy, counterterms, finite pieces, and higher-curvature contributions depend on field content and prescription. The report therefore avoids calling the bare coefficient universal.
 - Thermodynamic, entanglement, and microscopic interpretations are separated.** A finite effective-action entropy need not by itself identify the microscopic gravitational degrees of freedom being counted.
 - Physics v1.0 is preserved.** Its horizon-capacity condition remains exactly what the frozen source calls it: a structural domain extension. Gate 0 neither deletes it nor upgrades it to a derivation.

For any future route to license Gate 1, at least one new result must bridge the firewall. It must either (i) derive geometric/local-QFT structure for the original observer factor, (ii) introduce a new geometrically defined entropy variable and explicitly label that move as a new postulate, or (iii) derive/test a map between the generic observer entropy and a separately defined geometric entropy.

11 Stop declaration

Gate 1 is not entered. No modified Friedmann equation, effective information stress-energy tensor, cosmological perturbation equation, or gravitational-wave spectrum is derived from Gate 0. Such developments require an additional bridge not established in this report.

This stop declaration is methodological, not rhetorical. In particular, the phenomenological gravitational-wave ansatz of Physics v1.0 is not reverse-engineered into a first-principles prediction, and no effective stress-energy tensor is invented merely to continue the chain.



12 Future routes – not part of Gate 0

The following are branch points for future work only. None changes the frozen Gate 0 verdict.

Route I – preserve the v1.0 ontology

Keep

$$H_{\text{obs}} = -\text{Tr}(\rho_{\text{obs}} \log_2 \rho_{\text{obs}}) \quad (36)$$

as a generic observer/environment information variable. Accept that its horizon mapping is unresolved. This route makes no new geometric identification and therefore does not, by itself, license Gate 1.

Route II – introduce geometric structure

Define a distinct object, for example

$$H_{\Sigma} \equiv -\text{Tr}(\rho_{\Sigma} \log_2 \rho_{\Sigma}), \quad (37)$$

where Σ is a specified spatial/causal partition with a corresponding local field theory and regulator. This is a **NEW POSTULATE/DEFINITION**; it is not a derivation that the original H_{obs} was secretly geometric.

Route III – retain both variables and derive a map

Keep H_{obs} and H_{Σ} separate and investigate an explicit physical map

$$F : H_{\text{obs}} \longrightarrow H_{\Sigma} \quad (38)$$

or a more general relation $\mathcal{R}(H_{\text{obs}}, H_{\Sigma}; \theta) = 0$. Any such map must state its physical degrees of freedom, locality/geometry assumptions, dynamics, and falsifiable consequences. This route preserves the original ontology while exposing the missing bridge as the research object.

13 Conclusion

Physics v1.0 explicitly left the microphysical observer–horizon identification open. Gate 0 tested whether a specific body of established entanglement and semiclassical-gravity work closes that gap without changing the original assumptions. Srednicki and BKLS demonstrate area behavior in controlled geometric free/Gaussian reduced-state constructions. Susskind–Uglum and subsequent effective-action analyses show a genuine connection between matter entropy divergences and renormalization of gravitational couplings, including higher-curvature terms. Those results are physically substantive, but they operate on geometrically specified field-theory/horizon structures. The v1.0 observer entropy is defined by an abstract observer/environment factorization and contains no independently established geometric localization, regulated local-QFT UV structure, or gravitational effective-action coupling.

Accordingly,

$$\boxed{\text{GATE 0 VERDICT: UNRESOLVED IN GENERALITY}} \quad (39)$$

with the logically separate statement

$$\boxed{\text{UNRESOLVED} \neq \text{FAIL.}} \quad (40)$$

Gate 0 therefore freezes a gap rather than filling it by assumption. Routes I–III may investigate that gap in future work, but Gate 1 is not licensed from the present assumptions.



A Full research ledger

Entry	Tag	Ledger statement
0.1	[ASSUMPTION]	Physics v1.0 assumes $\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{obs}} \otimes \mathcal{H}_{\text{env}}$; it is not derived from Wheeler–DeWitt.
0.2	[ESTABLISHED]	Given a factorization and global state, $\rho_{\text{obs}} = \text{Tr}_{\text{env}} \rho$ is the standard reduced density operator.
0.3	[ESTABLISHED FROM v1.0]	$H_{\text{obs}} = -\text{Tr}(\rho_{\text{obs}} \log_2 \rho_{\text{obs}})$ is the bit-normalized von Neumann entropy associated with the chosen observer/environment correlations.
0.4	[OPEN]	$\mathcal{H}_{\text{obs}} \otimes \mathcal{H}_{\text{env}} \stackrel{?}{\leftrightarrow} \mathcal{H}_{\text{inside}} \otimes \mathcal{H}_{\text{outside}}$ is not established.
1A.1	[ESTABLISHED FROM LITERATURE]	Srednicki traces a free massless scalar vacuum across a spherical spatial partition and finds entropy proportional to area rather than volume.
1A.2	[SHARED ASSUMPTION]	Free-field ground state and linear partition preserve Gaussianity under partial trace, enabling covariance/spectral entropy calculation.
1A.3	[REGULATOR DEPENDENT]	In $3 + 1$ dimensions the leading bare term has schematic form $S \sim k_B C_S A / a^2$; the coefficient depends on UV prescription.
1A.4	[DERIVED LIMIT]	Setting $a \sim \ell_P$ only yields an order-of-magnitude A/ℓ_P^2 scale; it does not derive the Bekenstein–Hawking coefficient.
1B.1	[ESTABLISHED FROM LITERATURE]	BKLS associate a reduced density matrix with inaccessible black-hole-related field degrees and calculate an area-proportional contribution in a simplified Gaussian oscillator model.
1B.2	[MODEL DEPENDENT]	BKLS’s explicit calculation is not a universal continuum interacting-QFT theorem and does not derive Newton’s constant from entanglement.
1B.3	[DERIVED COMPARISON]	Srednicki and BKLS share local correlations, geometric division, partial trace, UV discretization, and Gaussian tractability; their surfaces and physical settings differ.
2.1	[ESTABLISHED FROM LITERATURE]	Susskind–Uglum find a quadratic near-horizon divergence for free-scalar entropy per area on a fixed black-hole background.
2.2	[ESTABLISHED FROM LITERATURE]	SU relate quantum entropy-per-area corrections to quantum corrections of the gravitational coupling.
2.3	[ESTABLISHED FROM LITERATURE]	Fursaev–Solodukhin show one-loop entropy divergences are removed by renormalization of G and higher-curvature couplings; higher-curvature terms contribute geometry-dependent entropy.
2.4	[FIELD/REGULATOR DEPENDENT]	Bare matter entropy depends on field content, couplings, and regulator; renormalization must be specified before interpreting a finite gravitational result.
3.1	[DEFINITIONAL SEPARATION]	$S_{\text{ent}}^{\text{matter}}$ is matter reduced-state entropy; S_{grav} is gravitational-action entropy; S_{gen} combines gravitational and exterior matter terms in semiclassical constructions.
3.2	[NOT DERIVED]	No dimensional or terminological argument establishes $S_{\text{ent}}^{\text{matter}} = S_{\text{grav}} = S_{\text{gen}}$.
3.3	[MODEL DEPENDENT]	Nonminimal coupling and higher-curvature terms complicate the entanglement/thermodynamic relation; the ontology is framework dependent.
4.1	[NOT ESTABLISHED FOR v1.0]	No geometric entangling/horizon surface is defined by the observer/environment labels alone.
4.2	[NOT ESTABLISHED FOR v1.0]	No local field content, UV regulator, or renormalization prescription is part of the H_{obs} definition.

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4.3	[NOT ESTABLISHED FOR v1.0]	No dependence of H_{obs} on G , $g_{\mu\nu}$, or higher-curvature effective-action couplings is derived.
4.4	[DERIVED DIAGNOSTIC]	Arrows 1–3 cannot be transferred across the Gate 0A/0B firewall under the original v1.0 assumptions.
4.5	[OPEN]	Additional theory could in principle supply the missing bridge; absence of such a bridge here is not a no-go theorem.
STOP	[FROZEN]	Gate 0 = UNRESOLVED IN GENERALITY. Gate 1 is not entered from current assumptions.

B Reference formulas and notation

For clarity, the principal quantities used in the audit are collected here:

$$\rho_A = \text{Tr}_{\bar{A}} \rho, \quad (41)$$

$$S_{\text{ent}}^{\text{matter}} = -k_B \text{Tr}(\rho_A \ln \rho_A), \quad (42)$$

$$H_{\text{obs}} = -\text{Tr}(\rho_{\text{obs}} \log_2 \rho_{\text{obs}}), \quad (43)$$

$$S_{\text{BH}} = \frac{k_B A_H}{4\ell_P^2}, \quad \ell_P^2 = \frac{\hbar G}{c^3}, \quad (44)$$

$$S_{\text{gen}} = S_{\text{grav}} + S_{\text{out}} \quad (\text{schematic semiclassical form}). \quad (45)$$

No equality between $k_B \ln 2 H_{\text{obs}}$ and any of the other entropy objects is assumed by this report.

References

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